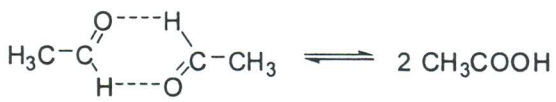
Physical Chemistry CUME
January 24, 2014
Announced Topic: Equilibria

1. Acetic acid exists mostly as a dimer in the gas phase. However, the dimer is partly dissociated according to the reaction



To determine the equilibrium constant and the enthalpy of dissociation of the dimer, a container of volume 100 cm^3 is filled with acetic acid vapor at 437 K and 1 bar pressure. After cooling, the acid condenses in the container. It is weighed and its total mass is 0.242 g. The experiment is repeated at the same pressure but at 471 K, and the total mass of the acid is found to be 0.187 g.

(A) Calculate the equilibrium constant for dissociation at these two temperatures,

$$K_P = (P_{acid}/P^o)^2/(P_{dimer}/P^o)$$

assuming that both acetic acid and its dimer behave like perfect gases (45 points).

(B) Calculate the enthalpy of dimerization from the van't Hoff equation,
$d \ln K/ d(1/T) = - \Delta_r H^o/R$ in kJ/mol (35 points).

(C) Is the equilibrium constant expected to increase or decrease as the temperature increases, and **why**? Does a positive enthalpy of dissociation imply that the dimer is less stable (higher in energy) than two monomers, or more stable (lower in energy)? (20 p)

The standard pressure P^o = 1 bar = 10^5 Pa, the gas constant in SI units is R=8.3145 J/(Kmol) = 0.083145 L bar K^{-1} mol^{-1}, the molecular mass of acetic acid is 60.05 g/mol.

(Hint: One can calculate the total number of moles in a 100 mL volume using the gas law. Assuming that the partial pressure of the dimer is P_{dimer}, we can calculate the number of moles and hence the mass of the dimer. The pressure of the monomeric acid is obviously P_{acid}=1 bar – P_{dimer}, and one can similarly calculate the mass of the monomeric acid in terms of the, as yet unknown, P_{dimer}. The sum of these is the total measured mass of acetic acid, dimerized or not. This can be solved for P_{dimer}. Once P_{dimer} and P_{acid} are known, the equilibrium constant can be calculated. The derivative d lnK/d(1/T) can be approximated by a finite difference, taking ln K and 1/T at both 437 and 471 K, forming the differences and dividing the difference in ln K by the difference in 1/T.)